EXAMINATION NUMBER_____

ANSWER PRIVATE ACADEMY

2019/20202 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATIONS

CHEMISTRY

September, 2018 FORM IV **Subject No: M022/II**

(100 Marks) **Time allowed: 2 hours**

PAPER I

INSTRUCTION:

1. This paper contains **15**pages. **Please check**.
2. Answer **all** questions in the blank spaces provided on the question paper. The maximum number of marks for each answer is indicated against each question.
3. Write your **examination number** on each page of this paper.
4. In the table provided on this page, **tick** against the question number you have answered.
5. Hand in your examination paper to the invigilator when time is called to stop writing.

Question number	Tick question 1 – 13 if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

SECTION A (70 marks)

Answer all the questions in this section

1a. Give any two branches of chemistry.

_____

_____

_____

_____ **(2 marks)**

b. What is the difference between "base units" and "derived units"

_____

_____

_____

_____ **(2 marks)**

c. Write down the symbols of any one examples of:

i. Base units

_____

_____(1 marks)

ii. Derived units

_____

_____(1 marks)

2a. Give any one type of polymerization

_____

_____

_____(1 mark)

b. Draw a diagram to show what happens when vinyl chloride molecules polymise to form polyvinylchloride.

(2 marks)

c. State any one way of minimizing wastes.

_____

_____

_____(1mark)

d. Give any two examples of greenhouse gases.

_____

_____

_____

_____(2 marks)

3. Below are the general formulae of some homologous series represented by letters, P, Q, R S and Z

P : $C_n H_{2n}$

Q : $C_n H_{2n+1} COOH$

R : $C_n H_{2n+1} OH$

S : $C_n H_{2n+1} CHO$

Z : $C_n H_{2n+1} CO C_n H_{2n+1}$

a. Name the homologous series represented by S and Z.

S _____

_____(1mark)

Z _____

_____(1 mark)

b. Draw the structure formula of a compound with 4 carbon atoms in homologous Q.

(2 marks)

c. Draw al the structural isomers of the compound with 4 carbon atoms in homologous series P.

(3 marks)

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d. Explain how a compound in homologous R could be distinguished from a compound S

_____

_____

_____

_____

_____(3 marks)

4. a. Define the term acid using Bronsted Lowry theory.

_____

_____

_____(1 mark)

b. Explain how local acid – base indicator can be made.

_____

_____

_____

_____

_____(4 marks)

c. Explain any two application of neutralization reaction in health

_____

_____

_____

_____

_____(2 marks)

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e. Ethanoic acid (CH_2COOH) reacts with water according to the following chemicals equation: $CH_3COOH(aq) + H_2O\ (l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$

i. Identify the two conjugate acid – base pairs

_____

_____

_____(2 marks)

ii. Explain how hydronium (H_3O^+) ions was formed.

_____

_____

_____

_____(3 marks)

5. a. Define water hardness.

_____

_____

_____**(1 mark)**

b. Name the ions that cause water hardness.

_____

_____

_____ **(2 marks)**

c. Explain any two ways of removing water hardness

_____

_____

_____

_____

_____(4 marks)

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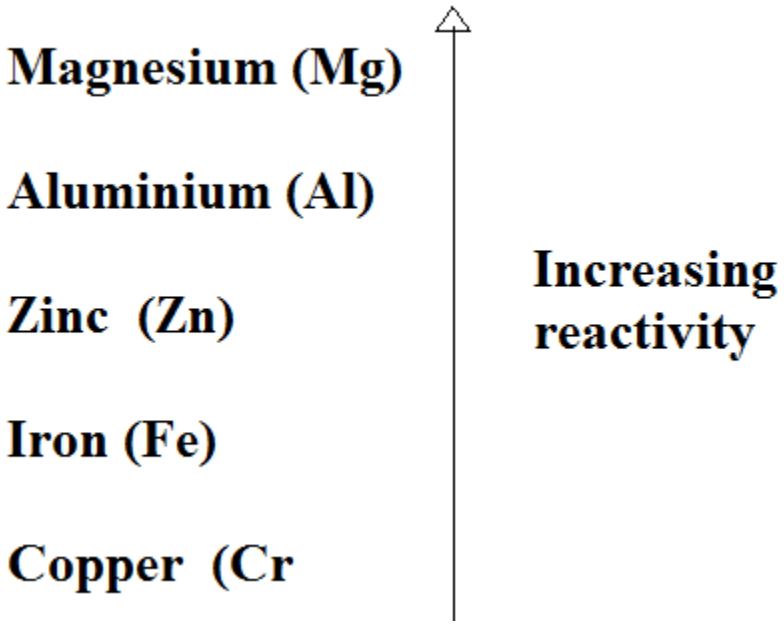
6. **a.** The following is part of a reactivity series.



i. Which two elements will displace Zinc from its Oxide.

_____

_____

_____(2 marks)

ii. Give a reason for the answer to 6a.

_____

_____

_____**(1 marks)**

b. The following are equations for the reaction between magnesium (Mg) and silver nitrate ($AgNO_3$):

$$2Ag^{+}(aq) + 2e \longrightarrow 2Ag^{0}(s)$$

$$Mg^{0}(s) - 2e \longrightarrow Mg^{2+}(aq)$$

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i. Identify the oxidation and reduction equation

Oxidation

_____

_____(1 mark)

Reduction

_____

_____(1 mark)

ii. Write a full chemical equation for the reaction.

_____

_____

_____(2 marks)

iii. Identify:

(1) An oxidizing agent

_____

_____

_____(1 mark)

(2) A reducing agent

_____

_____

_____(1 mark)

7. Figure 1 is a diagram showing nudei of two atoms.

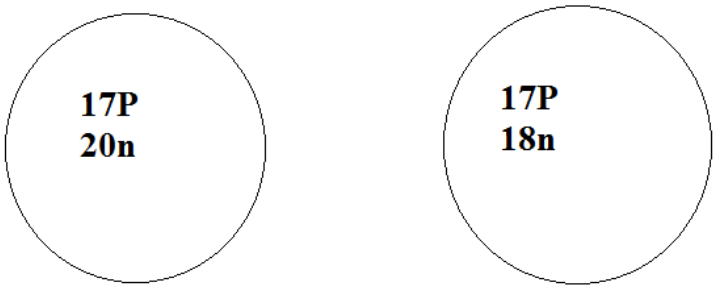

FIGURE 1

a. Explain why these atoms react in the same way.

_____

_____

_____

_____

_____ **(3 marks)**

b. What are atoms of this type called?

_____

_____**(1 mark)**

c. To which period of the periodic table could each belong?

_____

_____ **(1 mark)**

d. Explain the answer to 7c.

_____

_____

_____

_____ **(3 marks)**

8. **Table 1** shows the arrangement of some elements in the periodic Table.

H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K						Br	Kr

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i. To which period does Li belong?

_____

_____

_____**(2marks**

ii. Write the electron configuration for P.

_____

_____

_____**(1 mark)**

iii. Explain the trend in reactivity when moving up the group of **Br Cl** and **F.**

_____

_____

_____

_____**(4 marks)**

9. a. Define water of crystallization.

_____

_____

_____ **(1 mark)**

b. What do we call compounds containing water of crystallization

_____

_____

_____**(1 marks)**

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c. Calculate the percentage of water of crystallization in the chemical compounds with the formula:
$Na_2CO_3.\ 10H_2O$. RAM, H = 1, C + 12, O + 16 and Na = 23

(4 marks)

10. a. What is the difference between limiting reactant and excess reactant?

_____

_____

_____ **(2 marks)**

b. Magnesium and Oxygen react to form magnesium Oxide according to the equation;

$2Mg(s) + O_2(g) \longrightarrow 2MgO\ (s)$.

If 6g of magnesium and 2g of oxygen are available for the reaction.

i. Calculate the number of moles available for

(1) Magnesium

(2 marks)

(2) Oxygen

(2marks)

ii. Determine the limiting reactant and excess reactant

_____

_____

_____

_____**(2 marks)**

SECTION B (30 marks)

11. a. Define Frasch process.

_____

_____

_____(1 mark)

b. Using a well labeled diagram, explain how surphur is extracted from its underground deposits using the frasch process.

(9 marks)

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12. Explain how 500ml of 0.2m sodium chloride solution can be prepared using sodium chloride crystals. The explanation should include all the necessary mathematical calculation. (relative atomic mass of Na = 23 and Cl = 35).

_____

_____

_____

13. Ammonia is prepared industrially in a process called Haber process. With the aid of a well labeled diagram, describe how Haber process is used to produce ammonia (NH_3) from nitrogen (N_2) and hydrogen (H_2)

_____ **(10 marks)**

END OF QUESTION PAPER

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